

## Morphological decomposition in Austronesian voice: evidence for intermediary movement<sup>1</sup>

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**main takeaway:** Austronesian voice morphology—although seemingly in complementary distribution—should be attributed to the spell-out of distinct syntactic projections and their various combinations

- common syntactic analyses: voice morphology treated as a set of morphemes whose locus of spell-out is delegated to a single head (e.g. Rackowski 2002; Chen 2017; Erlewine et al. 2017, a.o.)

- often in complementary distribution, as below...

(1) Seediq [Taiwan; Atayalic]

- a. ga=[ku]      k(**m**)eruc rudux  
PROG=1SG.PIV <AV>cut    chicken  
'I am cutting chicken.'

**actor voice (AV):** infix <m>

- b. kret-**un**=mu    [ka rudux]  
cut-PV=1SG.GEN PIV chicken  
'I will cut *the* chicken.'

**patient voice (PV):** suffix -un

- c. s-keruc=mu    rudux [ka yayu nii]  
CV-cut=1SG.GEN chicken PIV knife this  
'I will cut chicken with *this* knife.'

**circumstantial voice (CV):** prefix s-

- NOTE: voice marker is **bolded**, pivot argument is in [brackets] and translated as *italics*

- **THIS TALK!** voice morphology is the result of several distinct functional projections...
  - closer examination reveals when they are bundled vs. come apart
  - previous literature: decomposition in Malagasy (Guilfoyle et al. 1992; Paul 1999; Pearson 2005; Travis 2010), applicative bundling in Formosan (J. Wu 2007; H. Chang 2009; Bogren Svensson 2024, a.o.), distinct historical origins (Starosta et al. 1982; Peterson 2017; Begus et al. 2023)

structural hierarchy: C > T > Neg > Asp<sub>PFV</sub> > **MT** > **E** > Asp<sub>RED</sub> > Voice > **Appl** > v > √ROOT

- functional projection that is present at default:
  - E(vent): event closure head, responsible for infinitive / (ir)realis marking (Travis 2010)
- functional projections that trigger “leapfrogging” movement / argument advancement:
  - M(iddle)T(opic): topic position at edge of extended Voice domain (LaCerde 2020)
  - Appl(licative): raising applicative head, reorders internal arguments (Georgala 2012)

- combines both insights from theories on Austronesian voice as argument structure vs. information structure alternations (Schachter 1976; Aldridge 2004; Rackowski & Richards 2005, a.o.)

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**why decompose?** if we take morphology as the starting point...

- [ORIENTATION] different orientation (prefix vs. suffix): morpheme-specific or structural difference?
- [AFFIX ORDER] morpheme and word order reflects clause structure under the Mirror Principle: voice morphology is located in the middle field, not high in the left periphery (Travis 2010, Wu 2024)
- [IRREALIS] irrealis voice paradigm is morphologically complex with containment relation (Wu 2025)
- [CLAUSE SIZE] voice markers have different syntactic distributions with respect to adverbs and complementation, forming an implicational hierarchy (cf. ICH, Wurmbrand & Lohninger 2023)

## 1 morphological evidence – from Seediq and other Formosan languages

### 1.1 affix and auxiliary order (Wu 2024)

under Baker's (1985) Mirror Principle, *morpheme order and syntax should reflect one another...*

morpheme order of verbal morphology  $\Leftrightarrow$  location of voice in clause structure

| Seediq VIs: |   |                               |                    | head which hosts it:     |
|-------------|---|-------------------------------|--------------------|--------------------------|
| PFV         | ⇔ | ⟨ <i>n</i> ⟩                  | infix location __V | Asp <sub>PFV</sub>       |
| AV          | ⇔ | ⟨ <i>m</i> ⟩                  | infix location __V | E                        |
| ANTIC       | ⇔ | V-                            |                    | Voice <sub>[-D]</sub>    |
| REC         | ⇔ | <i>p</i> -RED <sub>CV</sub> - |                    | Voice <sub>[+D]</sub>    |
| CV          | ⇔ | <i>s</i> -                    |                    | Appl (to be revised)     |
| CAUS        | ⇔ | <i>p</i> -                    |                    | <i>v</i> <sub>CAUS</sub> |

- **actor voice (av)** is located between Asp<sub>PFV</sub> and Voice, in functional projection E
  - based on order of left-aligned affixes (cf. infixes are underlyingly prefixes, Kalin 2022)

#1: Asp<sub>PFV</sub> > E, Appl      perfective PFV attaches outside of AV and CV

- (2) Bunun (NTU 2016):      *m* $\langle i \rangle$ -*ludah* '⟨PFV⟩AV-hit'      (vs. *l* $\langle in \rangle$ *udah* '⟨PV.PFV⟩hit')
- s* $\langle in \rangle$ -*pit'ia* '⟨PFV⟩CV-cook'      (vs. *p* $\langle in \rangle$ *it'ia* '⟨PV.PFV⟩cook')
- (3) a. Seediq:       $\langle m \rangle \langle n \rangle$ *imah* '⟨PFV⟩⟨AV⟩drink'      (vs.  $\langle n \rangle$ *imah* '⟨PV.PFV⟩drink')
- r* $\langle m \rangle \langle n \rangle$ *engo* '⟨PFV⟩⟨AV⟩talk'      (vs. *r* $\langle n \rangle$ *engo* '⟨PV.PFV⟩talk')

- b. AV  $\langle m \rangle$  and PFV  $\langle n \rangle$  both target the same infix location \_\_V:

*imah* 'drink'  $\rightarrow$   $\langle m \rangle$ *imah* '⟨AV⟩drink'  $\rightarrow$   $\langle m \rangle \langle n \rangle$ *imah* '⟨PFV⟩⟨AV⟩drink'  $\rightarrow$  *mnimah*

*rengo* 'talk'  $\rightarrow$  *r* $\langle m \rangle$ *engo* '⟨AV⟩talk'  $\rightarrow$  *r* $\langle m \rangle \langle n \rangle$ *engo* '⟨PFV⟩⟨AV⟩talk'  $\rightarrow$  *rmnengo*

#2: E > Voice      AV attaches outside of anticausative ANTIC (Chen 2020) and reciprocal REC

- (4) Puyuma (Chen 2020):      *d* $\langle em \rangle$ *isdis* '⟨AV⟩tear'       $\langle m \rangle$ *u-disdis* '⟨AV⟩ANTIC-tear'
- $\langle m \rangle$ *apit* '⟨AV⟩pile.up'       $\langle m \rangle$ *u-apit* '⟨AV⟩ANTIC-pile.up'
- (5) Seediq:      *t* $\langle m \rangle$ *akur* '⟨AV⟩trip'       $\langle m \rangle$ *[V]-takur* '⟨AV⟩ANTIC-trip' > '⟨AV⟩fall'
- d* $\langle m \rangle$ *engu* '⟨AV⟩roast'       $\langle m \rangle$ *[V]-dengu* '⟨AV⟩ANTIC-roast' > '⟨AV⟩dry'

- **circumstantial voice (cv)** is located between  $\text{Asp}_{\text{PFV}}$  and  $v$ , in functional projection Appl
  - more restrictive, does not co-occur with most Asp or Voice morphemes... more on this later!

#3: Appl >  $v_{\text{CAUS}}$  cv attaches outside of causative CAUS

- (6) Seediq: *p-qita* 'CAUS.AV-see' *s-p-qita* 'CV-CAUS-show'  
*p-beebu* 'CAUS.AV-hit' *s-p-beebu* 'CV-CAUS-hit'

**OBSERVATION #1:** voice morphology is located in the middle field, *not* in the left periphery (i.e. C/T)

- auxiliary and word order also suggests that **all verbal morphology is located below  $\text{Asp}_{\text{PFV}}$** 
  - C [*netun* 'if'] - T [*wada* 'PST', *maha* 'FUT'] - Neg [*ini* 'NEG'] - Asp [*ga* 'PROG'] - verb complex
  - tense and negative auxiliaries do not interfere with voice marking on the verb

- (7) a. *ga*=[ku] <**m**>ekan ido  
 PROG=1SG.PIV <AV>eat rice  
 'I am eating (rice).'
- c. *wada*=[ku] ini Ø-ekan ciga  
 PST=1SG.PIV NEG AV.IRR-eat yesterday  
 'I did not eat yesterday.'
- b. *ga*=mu puq-**un** [ka hei qhuni]  
 PROG=1SG.GEN eat-PV PIV fruit  
 'I am eating *the fruit*.'
- d. *wada*=mu ini puq-**i** [ka hlama]  
 PST=1SG.GEN NEG eat-PV.IRR PIV candy  
 'I did not eat *the candy*.'

**OBSERVATION #2:**  $\text{Asp}_{\text{PFV}}$  is the domain of word formation, within which voice morphology is located

## 1.2 irrealis voice paradigm (Wu 2025)

- **irrealis voice forms** are used in imperative, negative, and negative imperative constructions
  - no consistent "irrealis" marker, varies based on voice alternation

- (8) a. Ø-ekan ido!  
 AV.IRR-eat rice  
 'Eat (rice)!'
- b. puq-**i** [ka ido]!  
 eat-PV.IRR PIV rice  
 'Eat *the rice*!'
- c. biq-**an-i** yaku [ka patis]!  
 give-CV.IRR 1SG PIV book  
 'Give me *the book*!'

## (9) voice markers across Formosan (simplified)<sup>2</sup>

|         | INDICATIVE    |           |           |           | IRREALIS  |           |           |           |
|---------|---------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
|         | <u>AV</u>     | <u>PV</u> | <u>LV</u> | <u>CV</u> | <u>AV</u> | <u>PV</u> | <u>LV</u> | <u>CV</u> |
| Seediq: | < <i>m</i> >  | -un       | -an       | s-        | Ø         | -i        | -i        | -an-i     |
| Paiwan: | < <i>em</i> > | -in       | -an       | si-       | -u        | -u        | -i        | -an       |
| Bunun:  | <i>ma-</i>    | -un       | -an       | 'is-      | -a        | -a-v      | -a-v      | 'is- -a-v |
| Puyuma: | < <i>em</i> > | -a-u      | -a-i      | -an-a-i   | Ø         | -u        | -i        | -an-i     |
| Tsou:   | < <i>m</i> >  | -a        | -i        | -nen-i    | -         | -         | -         | -         |
| Amis:   | < <i>om</i> > | -en       | -an       | sa-       | Ø         | -en       | -         | sa- -en   |

<sup>2</sup>data on voice morphology is compiled from two sources: the 2018 grammar series on Formosan languages published by the Council of Indigenous Peoples, as well as the 2024 *Handbook on Formosan Languages* edited by Li, Zeitoun, & de Busser

**OBSERVATION #3:** voice morphology is *not* in complementary distribution in the imperative paradigm

- these forms are **internally complex** in many Formosan languages
  - containment relation in morphological complexity: AV (often null) < PV/LV < CV
  - if there is a containment relation, cv forms will always be the most morphologically complex!
- morpheme order has three main components:
  - call them 1, 2, and 3 for now... if AV is V-1, then PV is V-1-2, cv is V-3-1-2
  - voice morphemes in indicative vs. imperative: **bundled vs. decomposed** spell-out
- in other words, AV spells-out E, PV spells-out MT+E, and cv spells-out MT+E+Appl

|      |        |                     |            |            |              |
|------|--------|---------------------|------------|------------|--------------|
| (10) | Seediq | AV                  | PV         | LV         | CV           |
|      | IND    | $\langle m \rangle$ | <i>-un</i> | <i>-an</i> | <i>s-</i>    |
|      | IRR    | $\emptyset$         | <i>-i</i>  | <i>-i</i>  | <i>-an-i</i> |

|      |                   |                                       |                                          |
|------|-------------------|---------------------------------------|------------------------------------------|
| (11) | vocabulary items: | INDICATIVE                            | IRREALIS                                 |
|      | AV:               | $E \Leftrightarrow \langle m \rangle$ | $E \Leftrightarrow \emptyset / \_ [IRR]$ |
|      | PV:               | $MT+E \Leftrightarrow -un$            | $MT \Leftrightarrow -i / \_ [IRR]$       |
|      | CV:               | $MT+E+Appl \Leftrightarrow s-$        | $Appl \Leftrightarrow -an / \_ [IRR]$    |

**OBSERVATION #4:** voice can be decomposed into three distinct morphemes, with order V-Appl-E-MT

### 1.3 clause size and complementation

according to the Implicational Complementation Hierarchy (ICH) (Wurmbrand & Lohninger 2023), different clause types form *an implicational scale in terms of linguistic properties...*

- syntactic **distribution of voice markers** correspond to clausal complement size:
    - forms hierarchy from smaller to larger clauses: *Event* < *Situation* < *Proposition*
  - four levels of clause types:
    - cf. gerunds in Tagalog (Hsieh 2020), restructuring in Austronesian (Wurmbrand 2016, a.o.)
- |    |              |            |                                                                           |
|----|--------------|------------|---------------------------------------------------------------------------|
| #1 | gerund       | <b>vP</b>  | bare verb; no voice or aspectual morphology, no embedded structural agent |
| #2 | control      | <b>EP</b>  | AV-only; no non-AV or aspectual morphology, embedded agent is PRO         |
| #3 | <i>tough</i> | <b>MTP</b> | all voices; no aspectual morphology, embedded agent is PRO                |
| #4 | finite       | <b>CP</b>  | all voice and aspectual morphology possible, embedded agent is overt      |
- reflects position of voice projections in clause structure:  $[_{CP} C Asp [_{MTP} MT [_{EP} E Appl [_{vP} V ] ] ] ]$

**OBSERVATION #5:** different kinds of voice marking become available depending on clause size and height

- (12) a. prading  $[_{vP} \text{phuma}=\text{nami} \text{ paye npalic}]$   
 begin.AV plant=1PL.GEN rice glutinous  
 ‘Our planting of glutinous rice begins.’ gerund: bare verb

- b. prading=nami [EP ⟨m⟩huma paye npalic ]  
begin.AV=1PL.PIV ⟨AV⟩plant rice glutinous  
'We will begin to plant glutinous rice.' control: AV verb
- c. rriqu=mu [MTP slhay-**an** ka patis nii ]  
difficult.AV=1SG.GEN read-LV PIV book this  
'It is difficult for me to read *this book*.' tough: non-AV verb
- d. ⟨m⟩kela=ku [CP ⟨n⟩sa-**an**=na Awi ka Plsiya ]  
⟨AV⟩know=1SG.PIV ⟨PFV⟩go-LV=3SG.GEN Awi PIV Puli  
'I know that Awi went to *Puli*.' finite: finite verb

- syntactic distribution of voice markers also corresponds to **adverbial hierarchy** (Ernst 2001):
  - cf. adverbial verbs in Formosan (Chang 2009; 2010; Bogren Svenson 2024)
- adverbs have verb-like properties (cf. lexical vs. functional restructuring, Wurmbrand 2004)
  - event-oriented (all voices) < aspectual (AV and PV only) < speaker-oriented (bare adverb)
  - low to high: starting point of head movement through subsequent functional heads

[<sub>CP</sub> C *speaker-oriented* [<sub>MTP</sub> MT [<sub>EP</sub> E *aspectual* Appl [<sub>VP</sub> *event-oriented* V ] ] ] ]

- (13) a. wada=[ku]=na s-gruus m-osa Plsiya  
PST=1SG.PIV=3SG.GEN CV-secretly AV-go Puli  
'He secretly went to Puli for *me*.' event-oriented: [MT-E-AppI-Adv] [V]
- b. kntte-**un**=mu m-osa [ka Plsiya]  
often-PV=1SG.GEN AV-go PIV Puli  
'I often go to *Puli*.' aspectual: [MT-E-Adv] [AppI-V]
- c. yaani=[ku] m-osa Plsiya  
possible=1SG.PIV AV-go Puli  
'I will possibly go to Puli.' speaker-oriented: [Adv] [MT-E-AppI-V]

#### 1.4 summary

structural hierarchy: C > T > Neg > Asp<sub>PFV</sub> > **MT** > **E** > Asp<sub>RED</sub> > Voice > **AppI** > v > √ROOT

#1 voice morphology is located in the middle field, within the domain of word formation Asp<sub>PFV</sub>

#2 voice morphology is decomposable and involves three different heads

#3 a containment relation AV < PV < CV holds in terms of morphological and structural complexity

*in the next section, we will discuss the syntactic function of the three projections MT, E, and AppI...*

## 2 syntactic analysis

- **Austronesian voice systems** are debated and controversial (Guilfoyle et al. 1992, Rackowski 2002, Aldridge 2004, Pearson 2005, Rackowski & Richard 2005, Chen 2017, Erlewine et al. 2017, a.o.)
  - a range of analyses: case, agreement, extraction, valency change, object shift, topic, etc.

- *however*, there is a descriptive consensus that can be framed in terms of **argument promotion**...
  - AV promotes the subject, PV promotes the object, and CV promotes the oblique
  - in Relational Grammar (RG) terms, these are arguments 1 vs. 2 vs. 3

|              | AV $\langle m \rangle$ | PV/LV <i>-un/-an</i> | CV <i>s-</i>   |
|--------------|------------------------|----------------------|----------------|
| intransitive | subject                | -                    | <b>oblique</b> |
| transitive   | <i>agent</i>           | <u>theme</u>         | <b>oblique</b> |
| ditransitive | <i>agent</i>           | <b>goal</b>          | <u>theme</u>   |
| causative    | <i>causer</i>          | <b>causee</b>        | <u>theme</u>   |

- previous theories of voice alternations did not account for complex morphological behavior...
  - case or agreement based approaches:
    - \* Rackowski 2002: nominative vs. accusative vs. oblique case
    - \* Chen 2017: subject agreement  $T_{[u\phi]}$  vs. object agreement Voice<sub>[F]</sub> vs. elsewhere case
  - argument structure and applicative based approaches:
    - \* Aldridge 2004: intransitive  $v_{INTR}$  vs. transitive  $v_{TR}$  vs. transitive  $v_{TR}$  w/ Appl
    - \* Rackowski & Richards 2005: no shift vs. object shift  $v_{[EPP]}$  vs. object shift  $v_{[EPP]}$  w/ Appl
- given a **morphology-to-syntax mapping**, complexity in one should be reflected in the other

**THIS ANALYSIS!** making explicit the functional projections involved to capture existing insights

## 2.1 functional projections

- expands on analysis of voice morphology in Malagasy (Travis 1991, 1994, 2010, Pearson 2005)
  - also decomposed into three parts: Asp (assigns ACC), E (assigns NOM), Appl
  - but not adopting case or spell-out filter analysis, and positing additional projection MT

*within a Relational Grammar framework*... MT and Appl represent **argument advancement rules**

(i) E is the default (ii) MT is 2 over 1 (ii) Appl is 3 over 2

AV: E (1)      PV: MT and E (2-to-1)      CV: Appl and MT and E (3-to-2-to-1)

- argument advancement rules have to be stepwise (cf. Kinyarwanda, Kimenyi 1976)

structural hierarchy:  $C > T > Neg > Asp_{PFV} > \mathbf{MT} > \mathbf{E} > Asp_{RED} > Voice > \mathbf{Appl} > v > \sqrt{ROOT}$

- functional projection that is present at default:
  - **E**(vent): event closure head, responsible for infinitive / (ir)realis marking (Travis 2010)
- functional projections that trigger “passive” movement / argument advancement:
  - **M**(iddle)**T**(opic): topic position at edge of extended Voice domain (LaCerde 2020)
  - **Appl**(icative): raising applicative head, reorders internal arguments (Georgala 2012)

### 2.1.1 *E(vent)*

#1 **E(vent)**: event closure head, responsible for **infinitive** / **(ir)realis marking** (Travis 2010)

- in Davidsonian event semantics, existential closure  $\exists$  that binds the event argument  $e$  of verb
- cf. “default av” in restructuring (Wurmbrand 2016; Chang 2017) is akin to infinitival form

- (14) a. wada=[ku] c<**mm
 PST=1SG.PIV <AV>forget <AV>eat medicine  
 ‘I forgot to take medicine.’ matrix subject is pivot: [V-AV V-AV]**
- b. wada=mu chngiy-**un** <**m
 PST=1SG.GEN forget-PV <AV>eat PIV medicine  
 ‘I forgot to take *the* medicine.’ embedded object is pivot: [V-PV V-AV]**

- see earlier data on indicative vs. irrealis voice paradigms: E as realis [REAL] vs. irrealis [IRR]

- evidence from other languages also suggests an intermediate position between T and V:
  - French: non-finite verbs appear after *pas* and before adverbs, finite verbs appear before both
  - literature on subjunctives (Rivero & Terzi 1995; Cinque 1999; Zanuttini et al. 2012, a.o.)

### 2.1.2 *M(iddle)Topic and Appl(icative)*

*these are syntactic projections that serve as morphological reflexes of argument advancement...*

- cf. “leapfrogging” in object shift (Bobaljik 1995; Rackowski & Richards 2005)
- specifically, it appears to violate locality by moving not the highest but the second highest

#2 **M(iddle)T(opic)**: topic position at edge of the extended Voice domain (LaCerde 2020)

- two places for information structure: 1) left periphery, 2) middle field
- see earlier data on voice morphology located within the domain of word formation  $Asp_{PFV}$

- evidence from other languages also suggests the existence of **middle field topics**:
  - Brazilian Portuguese: topic constituents can appear sentence initially or medially
  - sentence-internal topics in Chinese (Paul 2002) and “low-IP” topics in Italian (Belletti 2004)

(15) Brazilian Portuguese (LaCerde 2020)

- a. [na festa de Natal]<sub>TOP</sub> eu dei dois livros pra Maria  
 in.the party of Christmas I gave two books to.the Mary  
 ‘I gave two books to Mary at the Christmas party.’ left-periphery topic
- b. eu dei [na festa de Natal]<sub>TOP</sub> dois livros pra Maria  
 I gave in.the party of Christmas two books to.the Mary  
 ‘I gave two books to Mary at the Christmas party.’ middle field topic

#3 **Appl(icative)**: raising applicative head, reorders internal arguments (Georgala 2012)

- two types of Appl: 1) **thematic**—introduces arguments, 2) **expletive**—moves arguments
- cf. applicative analysis of cv (Rackowski 2002; Aldridge 2004; Chang 2009; Nie 2019)

- see data below on animate themes that raise into Spec,Appl, causing word order effects

- (16) a. pi-imah=[ku] *begu laqi*  
CAUS.AV-drink=1SG.PIV soup child  
'I made the child drink soup.' inanimate theme: [*theme* - **causee**]
- b. pe-emux=[ku] **Mahung laqi**  
CAUS.AV-hug=1SG.PIV Mahung child  
'I let Mahung hug the child.' animate theme: [**causee** - *theme*]

- evidence from other languages also suggests the existence of **internal reordering**:
  - German: direct objects and indirect objects can interchangeably have DAT or ACC case
  - literature on ditransitives (Larson 1988| Georgala 2012) and possessor raising (Deal 2013)

*in other terms, these projections can be thought of as two “passives”: an outer passive and an inner passive...*

- passives in English have two components: 1) agent demotion, 2) patient promotion
- typically seen as the former feeding the latter, but perhaps they are independent operations

## 2.2 derivation

we are ready to derive voice alternations in Austronesian! *some starting assumptions...*

**#1** nominative-accusative case alignment (Chen 2023) that is separate from pivot movement

- (17) a. p-pyas-**an** na laqi [ka sapah=mu]  
RED-sing-LV GEN child PIV house=1SG.GEN  
'The children will sing in my house.' (Chen 2023) unergative agent has GEN *na*
- b. h-hqil-**an** na riso nii [ka Paran]  
RED-die-LV GEN young.man this PIV Paran  
'This young man will die in Paran.' (Chen 2023) unaccusative theme has GEN *na*

**#2** binding relation is  $1 > 2 > 3$  (Chen 2017), reflects base generated position of arguments

- (18) a. wada=mu biq-**un** patis= $na_i$  [ka Mahung $_i$ ]  
PST=1SG.GEN give-PV book=3SG.GEN PIV Mahung  
'I gave *Mahung* her book.' IO is pivot: IO > DO
- b. wada=mu **s**-bege Mahung $_i$  [ka patis= $na_i$ ]  
PST=1SG.GEN CV-give Mahung PIV book=3SG.GEN  
'I gave Mahung *her* book.' DO is pivot: IO > DO

**#3** topical nature of pivot (Schachter 1976), topics create new binding relations (Chierchia 2024)

- (19) a. \*wada **m**-emux kn-kingal laqi $_i$  [ka bubu= $daha_i$ ]  
PST AV-hug RED-one child PIV mother=3PL.GEN  
intended: 'Their mother hugged every child.' AG is pivot: WCO violation
- b. wada mex-**un** na bubu= $daha_i$  [ka kn-kingal laqi $_i$ ]  
PST hug-PV GEN mother=3PL.GEN PIV RED-one child  
'Their mother hugged *every* child.' TH is pivot: WCO obviated

**#4** predicate fronting of vP and TP derives pivot-final word order (Massam 2001, Aldridge 2004)

let's see how this works with a simple ditransitive sentence...

- **the setup:**

1. pivot argument is merged in base location as a DP with topic feature [TOP]
2. MT/Apl probe for goal with [TOP] feature, and moves the goal into its specifier
3. C probes for the highest DP, and moves the DP into its specifier

- E is always present, MT and Apl are projected when they create an **interpretational difference**

|            |                              |                              |
|------------|------------------------------|------------------------------|
| <b>AV:</b> | E is projected               | MT and Apl are not projected |
| <b>PV:</b> | E and MT are projected       | Apl is not projected         |
| <b>CV:</b> | E, MT, and Apl are projected |                              |

- cf. Travis 1984 on Germanic V2 movement— subjects raise to Spec,TP but not Spec,CP
- MT and Apl serve the function of rearranging which argument is the highest, otherwise their presence is vacuous, e.g. if MT were projected in an AV derivation, it would still attract  $DP_{AG}$
- probing for [TOP] feature circumvents locality (cf. van Urk 2015)

- E, MT+E, and MT+E+Apl are spelled-out as **spans** (Svenonius 2016; 2020)

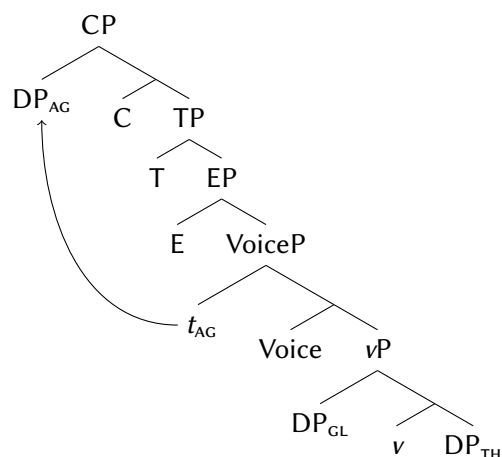
(20) AV construction:

$DP_{AG}$  has [TOP] feature

C finds closest DP

→  $DP_{AG}$  moves to Spec,CP

AV:  $\langle m \rangle \leftrightarrow E$



(21) PV construction:

$DP_{GL}$  has [TOP] feature

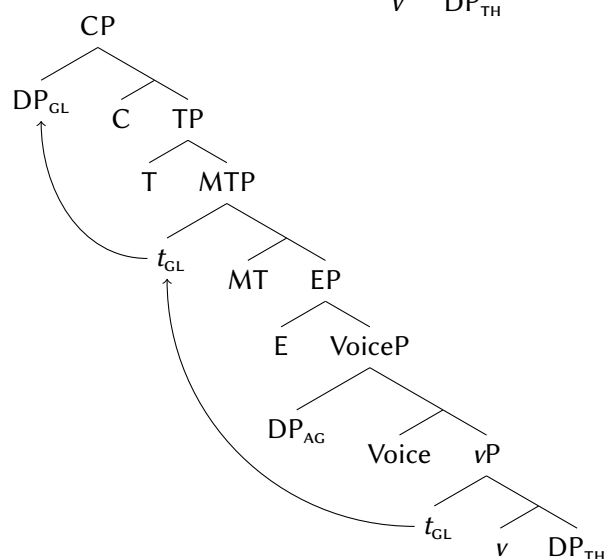
MT finds  $DP_{GL}$  w/ [TOP]

→  $DP_{GL}$  moves to Spec,MTP

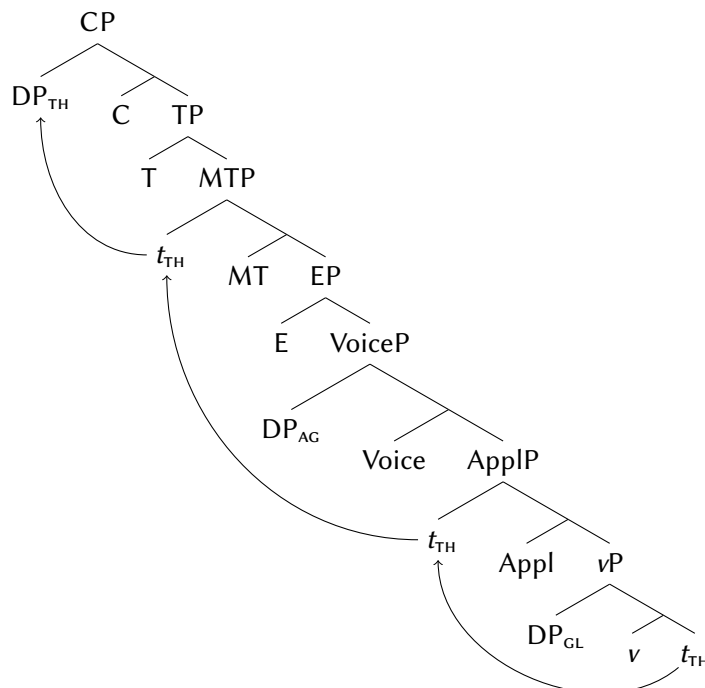
C finds closest DP

→  $DP_{GL}$  moves to Spec,CP

PV:  $-un \leftrightarrow MT+E$



- (22) cv construction:
- DP<sub>TH</sub> has [TOP] feature
- Appl finds DP<sub>TH</sub> w/ [TOP]  
 → DP<sub>TH</sub> moves to Spec,ApplP
- MT finds DP<sub>TH</sub> w/ [TOP]  
 → DP<sub>TH</sub> moves to Spec,MTP
- C finds closest DP  
 → DP<sub>GL</sub> moves to Spec,CP
- cv: si- ↔ MT+E+Appl



revisiting earlier morphological generalization, we can now explain them in syntactic terms...

structural hierarchy: C > T > Neg > Asp<sub>PfV</sub> > **MT** > **E** > Asp<sub>RED</sub> > Voice > **Appl** > v > √ROOT

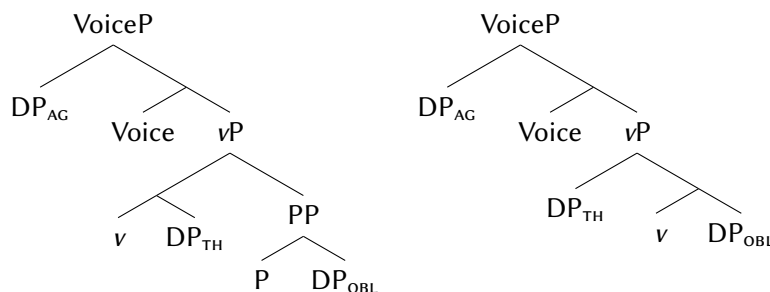
- #1 voice morphology is located in the middle field, within the domain of word formation Asp<sub>PfV</sub>  
 ↪ MT, E, and Appl are located below Asp<sub>PfV</sub> in the structural hierarchy
- #2 voice morphology is decomposable and involves three different morphemes  
 ↪ these are MT, E, and Appl; the difference between voice morphology being in complementary distribution or decomposed is due to MT, E, and Appl spelling-out as spans vs. individually
- #3 a containment relation AV < PV < CV holds in terms of morphological and structural complexity  
 ↪ AV projects E; PV projects MT and E; CV projects MT, E, and Appl

## 2.3 potential issues

- can this account easily extend to (in)transitive sentences with oblique arguments?
  - two potential structures for oblique argument below, will assume the righthand structure
  - binding relations invariable despite voice alternations (Chen 2017): AG > TH > OBL

## (23) transitive w/ oblique

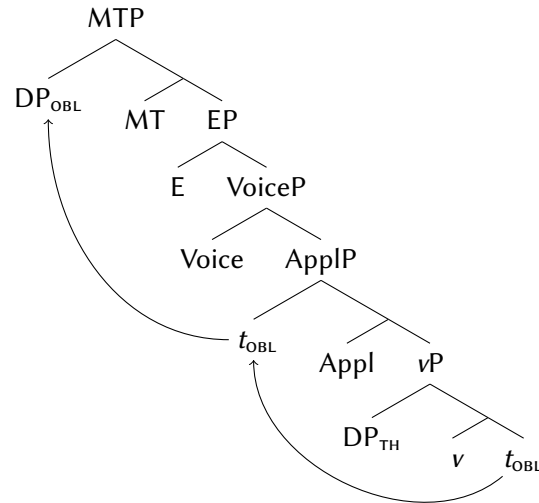
Av: agent  
 pv: theme  
 cv: oblique



- we assumed that MT and Appl are only projected when they create an interpretational difference
  - ...*but* MT is vacuous in unaccusatives, and Appl is vacuous in unergatives
  - tentative solution: intransitives are underlyingly transitive? (cf. Hale & Keyser 1993)

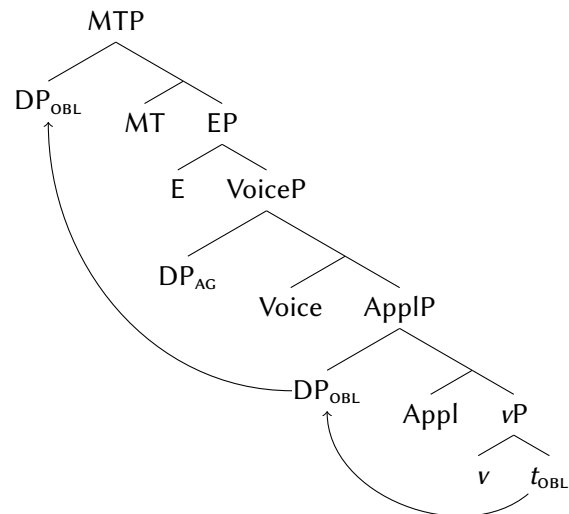
(24) unaccusative w/ oblique

AV: theme  
CV: oblique



(25) unergative w/ oblique

AV: agent  
CV: oblique



- can this account extend to languages where voice alternations affect binding relations?
  - cf. cv pivots in Tagalog cannot be bound by other internal arguments (Rackowski 2002)
  - tentative solution: difference in Appl as thematic vs. expletive applicative (Georgala 2012)

### 3 conclusion

**main takeaway:** Austronesian voice morphology—although seemingly in complementary distribution—should be attributed to the spell-out of distinct syntactic projections and their various combinations

- *understanding morphology leads us to reevaluate our syntactic analyses of Austronesian voice*
- these language appear to have **stepwise argument advancement**, as opposed to long movement

- open questions:
  - why is intermediate movement needed? vs. one long movement to Spec,CP? (cyclic phases?)
  - why are there different positions in the clause for topic movement? (discourse templates?)

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